

Global Governance @ Walmart, AI/ML Northstar Vision

TL;DR -- 30% cut in risk-adjusted loss ($\approx \$XXXM/yr$) by **unifying Global Governance** (Physical, Digital, Financial, Legal, Supply-Chain & ESG, Cyber/Workforce) onto one **continuously improving AI platform**. **Crawl -> Walk -> Run** roadmap delivers value inside **24 months**, **upskills talent** and **exits high-cost vendors**.

Where we sit (H2 2025 snapshot)

Ranked No. 1 on the Fortune 500, Walmart is a global-scale giant nearing 1T in market capitalization -- and at Walmart scale, all risks are amplified. Global Governance sits centrally and acts as Walmart's immune system; derisking the existing business and unlocking future upside via compliant, international growth.

Over time, with growing complexity and aging systems, Global Governance itself faces internal execution risks; slowdowns in employee productivity, ballooning operating costs, and business demands outpacing delivery. Meanwhile, socioeconomic trends have led to a rise in vigilantism (e.g., 2024 United Healthcare shooting), increasing global security requirements across the workforce. Alongside, tough macroeconomic headwinds (high interest rates, abrupt tariff decisions, debt spiral and limited debt monetization options) limit the effectiveness and feasibility of traditional scale-through-headcount growth options (i.e., reduction in effective-ROIC on headcount spend).

Machine Learning sits perfectly positioned to help address these issues.

This paper focuses on a strategic vision for ML (and AI) for Global Governance at Walmart:

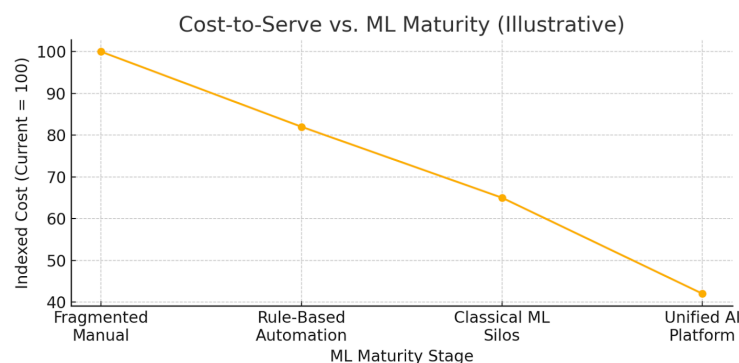
- [Inefficiencies and Opportunities](#)
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Inefficiencies and Opportunities

- 1) **Inefficiency:** There are **too many fragmented systems** each incompletely solving overlapping problems (e.g., document management, workflow orchestration and execution, audit logging and data streams, entity detection from multi-modal streams, escalation and tiered fallback processes, etc.).
 - a) **Opportunity:** Invest in AI-first platforms across the stack (problem definition, selection, evaluation, training and serving, deployment and operationalization) and eliminate or consolidate point-solutions.
- 2) **Inefficiency:** The **organizational complexity of getting-things-done is too high**; the resultant team output is too low to meet business needs (e.g., meaningful changes can not be shipped whole-hog, end-to-end by one team in less than 1 quarter -- everything needs x-team, x-function coordination).
 - a) **Opportunity:** Reorganize Global Governance teams into AI-first pods owning end-to-end business domain ownership. Empower hybrid teams (product owners, business leads, development teams) to own vertical solutions with each system developed as part of the broader ML platform umbrella; solving x-team ML issues and benefitting the entire org.
- 3) **Inefficiency:** The **cost of doing business is rising** and the risks of business non-compliance with regulations are increasing (e.g., fines for out-of-compliance product tariffs, unknown or missed regulations in net-new Geos).

- a) **Opportunity:** Encode all business workflows as evented Workflow-as-Code processes, generate and centralize rule and policy engines; enabling AI-driven operations and reducing costly mistakes.
- 4) **Inefficiency:** Head-count scaling is sub-linear; **platform leverage is required** (i.e., system complexity is outpacing operational capacity).
 - a) **Opportunity:** Allocate opex budget (> 10M/year) to drive all workloads onto AI platforms, develop systems ML-first by default. Standardise patterns and upskill teams on one reference architecture with concrete AI-first design patterns and strong forward-looking technical leadership.
 - b) **Crawl-Walk-Run** phased execution path:
 - i) **Crawl:** consolidate siloed data into a single event bus and stand-up “rules + classical ML” baselines for the top-three pain-points within **2 quarters**, POC first 5 agentic workflows via centralized Agentic AI platform.
 - ii) **Walk:** extend the platform to *all* Governance domains, introduce neural / multimodal models and edge distillation, and migrate **≥ 60 %** of workflows by **month 18**.
 - iii) **Run:** switch to a mixture-of-experts core, continuous reinforcement learning, and auto-allocation of prevention spend versus residual risk, achieving the **30 % risk-loss delta** before month 24.

ML Framework: Towards a single Global Governance AI



Goal: a single, unified AI-based governance platform that safeguards customers, employees, sellers and the enterprise, while accelerating compliant growth in every market. To achieve this vision, **we need a single objective function for Global Governance ML models**, generic enough to optimize over all Global Governance use cases without overfitting.

- **Modeling:** Governance ultimately optimizes **expected risk-adjusted loss** under resource constraints.
 - The probability-weighted financial or reputational impact of every event across Global Governance domains + a tunable cost term. Expressing risk this way condenses all heterogeneous signals (e.g., dangerous object detection video models, escalate-or-ignore AML-models) into one optimisation target, enabling a single MoE-style “WalmartAI” to continuously learn company-wide policies while still honouring domain-specific constraints.
- **Architecture:** Signals -> AI -> Policy -> Action, orchestrated in tight **predict-act-feedback** loops.
 - **Signals:** Realtime ingestion of all multi-modal data and business workflows into shared-state system-of-record and common embedder and encoder language.

- **AI:** Mixture-of-Experts core with domain adapters; continuous fine-tuning on real-world feedback. Model freezing and clear lineage for bias and drift mitigation. Distillation and edge-serving for cost/privacy/latency tradeoff options.
- **Policy:** Declarative optimizer that trades incremental prevention spend against residual risk; outputs guardrail settings and observes compliance/regulatory rules.
- **Action:** Respond and remediate across *Physical, Digital, Financial, Legal, Supply-Chain & ESG, Cyber/Workforce* domains with automatic, human-in-the-loop fallback and escalation.
- **Minimal-Viable-Model Framework:** Select the “lowest-rung” on a predefined ML-model decision matrix that solves for risk tolerance/SLAs; higher-rungs need justification (modality, latency, explainability, latency, privacy, or cost requirements).
 - **Rungs:**
 - Deterministic Rules/SQL
 - Low-risk & ambiguity, low cost to deploy and maintain, least powerful
 - Classical ML (GBDT, regressions, etc.)
 - Rich, structured, tabular
 - Neural Networks
 - Quantized
 - Edge, constrained resources
 - True Deep Learning
 - Highest complexity problems describable as a “single” question to answer
 - Agentic AI
 - Multi-step and global optimizations; asynchronous workflow automation
 - **Decision Dimensions:**
 - **Modality** (Single-Objective Supervised to “full” Multimodal) -- select least data-bandwidth intensive modality inclusive of input data types.
 - **Latency** (Batch vs. Realtime, hard vs. soft requirements) -- select for the maximum acceptable hard latency requirements.
 - **Explainability** (Human-in-the-Loop and understandability) -- select inversely proportional to context volume need to correctly intervene (e.g. “How much context does the human {queue worker/rater} need to correctly intervene?”).
 - **Regulation and Compliance** -- specify at problem definition stage (before data link between ruleset and data source persisted) to avoid risk and remain globally compliant.
 - **Cost and Privacy** -- Distill and deploy domain-specific requirements into cheaper edge-served models for latency/cost and privacy tradeoffs.

Success Criteria

- **Culture:** Walmart regularly speaks at major AI conferences on “Enterprise AI Deployments”; acts as a strong proxy and leading indicator for how Walmart’s tech team is perceived externally.
- **Vendor Exit-List:** Each vendor dependency has clear EOL milestones (time, operating and maintenance cost threshold) saving 100M/annually.
- **Public Goodwill:** 3rd party sourced public sentiment for *Walmart Ethics and Safety* shifts 1-stddev “to the right” YoY.

- **Governance Leading from the Front:** Within Walmart Tech, Global Governance seen as the frontrunners for applied AI; measured in % of x-org usecases.

Notes and Applied Decision Making Framework

- **All data is training data.** Deleting business data that we've generated is only permitted after synthesizing (extracting its value) into a model or feature store. To comply with legal erasure, we can use a write-once, per-region encrypted blob with cryptographic tombstones for deleted subjects. All business processes must be encoded as Workflows-as-Code (e.g., Infrastructure-as-Code evolution).
- **Rule changes require differential retraining.** New data must update our priors; balance cost/sensitivity tradeoffs per domain via central governing body for applied AI -- this builds expertise in trading off for business constraints and generates golden dataset to automate.
- **Transparency and explainability accelerate our flywheel.** Enable employees and users of AI systems to **ask any question**. Only two states can emerge from a question: an answer generated by Global Governance AI from knowledge graph, or an edit to the graph itself.
- **Agentic Workflows by-default.** Simplify and consolidate service architecture into a modern, serverless architecture using the agentic workflow design pattern (multi-modal reasoning agent as core runner), enabling arbitrary data connections via standard protocols (e.g., local and remote MCPs), using standard auth and control planes to operate.

Addendum -- Technical Deep-Dive

1. Mixture-of-Experts (MoE) Core -- Training & Routing Specifics

Feedback: *MoE Nitty-Gritty: The paper talks about using a "Mixture-of-Experts (MoE) core" for the AI. But it doesn't really explain how those different expert models will be trained or how the system decides which expert to use for what task. That's a big part of how it works!*

Ziyad's Response:

MoE/Mixture-of-Experts

Core architecture has ~**30 expert cells**: six governance domains (Physical-Security, Digital-Fraud/AML, Cyber & Workforce, Supply-Chain & ESG, Legal, Product-Safety) and five data modalities (Vision, Text, Tabular, Graph, Time-series). Each cell shares a frozen, cross-modal backbone (T5-XXL + ViT-G) but **unfreezes the final two Transformer blocks and adds a LoRA head** so we can specialise rapidly without forgetting. Training runs in three passes:

1. **Backbone pre-training** on Walmart's full governance telemetry (self-supervised + multi-modal contrastive).
 - a. Self-Supervised Backbone Pre-training on 4B multi-modal events (text, CCTV, POS, audit logs, sensor graphs).
 - b. Objectives: span-corruption (T5), masked-patch (ViT), mCPC (multi-modal Contrastive Predictive Coding).
 - c. Optimiser: AdamW ($\beta_2 = 0.95$), LR = $1e-4$ with cosine decay, ZeRO-3 sharding.
2. **Cell fine-tuning** on domain-specific labelled events using a unified risk-adjusted-loss; rule-based labels bootstrap data-sparse slices. The process is continuous, aligning with the "Run"-phase mandate to swap in an MoE core with reinforcement signals.
 - a. Unified risk-adjusted loss with bootstrapped labels: heuristics -> weak-label aggregation -> active-learning queries
 - b. Continual updates via parameter-efficient LoRA – no backbone gradient checkpointing, enabling sub-5 min SLO for hot-patches
3. **Periodic distillation** compresses the merged policy back to edge replicas (~<50 ms store latency) while registry hooks enforce lineage, bias, and drift gates -- the same "domain-adapter + continuous fine-tuning" loop.

Stale experts auto-queue for incremental refresh whenever their risk-delta or data distribution breaches SLOs.

Model Routing

A 2-trillion-parameter backbone is costly; routing ensures only the 1-2 most relevant experts run per token, hitting the cost-to-serve curve the vision document targets.

Two-Stage Gate:

- **Modality Gate:** Embedding norms + sensor meta-data; skips whole towers if input lacks modality via Top-k softmax over modality logits, load-balanced by FLOP-penalty
- **Domain Gate:** Risk features (entity-type, geography, SKU, channel) chooses the domain cell whose priors minimise *expected* residual loss via Context-aware MLP trained with GShard-style sparsity + auxiliary load-balancing loss

The composite router is **token-level** but throttled by a per-request FLOP budget, preventing the AML pipeline from hogging GPU when, say, a store camera sends a hundred low-risk frames.

Router weights are updated alongside experts with a **mixture-of-experts load-balancing loss plus a latency regulariser**. Reinforcement signals come from the same risk-adjusted objective that powers expert fine-tuning, closing the predict--act--feedback loop highlighted in the original architecture diagram.

2. Continuous Reinforcement Learning (RL) Loop -- Algorithms, Rewards & Environment

Feedback: "RL Algorithms: It mentions "continuous reinforcement learning" for the advanced 'Run' phase. We don't get details on which specific algorithms they're thinking of, or what the 'rewards' and 'environment' would look like for this governance AI to learn from."

Ziyad's Response:

RL Algorithms

- **Warm-Start:** Offline CQL (Conservative Q-Learning on last half-year of historical decision) provides safe initial policy and avoids reward hacking from zero-shot actor-critic adversarial setups
- **Simulated Pre-Training:** Importance Weighted Actor-Learner Architectures (IMPALA) + V-trace (distributed actor-critic) inside a large Ray RRLib cluster provides scales to trillion steps/day with recurrent head and supports long temporal credit assignments
- **Online Fine-tuning:** R2D2 (Recurrent DQN) CPO (Constrained-Policy Optimization) enforces hard SLA constraints and reduces legal-violation probabilities to acceptable tradeoff thresholds ($< 1e^{-6}$) and is data efficient for spiky event rates to stay within guardrails.

RL Environment

- **States:** 15k-dimension vector from the MoE's fused embeddings (Vision + Text + Tabular + Graph + Time-series) X curated risk features (e.g., entity_age, sku_margin, CCTV crowd-density) guarantees partial observability is minimised; no raw PII leaves the sandbox.
- **Actions:** Discrete-set of hierarchically grouped: *Hard fences* -- block TX, lock account, recall SKU and *Soft controls* -- require MFA, increase audit sample rate, *Comms* -- human escalation, Slack push, Jira ticket mirrors real governance levers; keeps effective latency ≤ 500 ms
- **Transitions:** Real-time event stream *digital-twin* simulator allows billions of off-policy updates/day without touching prod
- **Rewards:** $fn = - (A \times \text{Residual Risk}) - (B \times \text{Opportunity Cost}) - (C \times \text{Regulation Penalty})$ aligns with risk-adjusted EBITDA single objective

3. Real-Time Multi-Modal Data Pipeline and Bias & Drift Mitigation Playbook

Feedback: "Data Pipeline Details: While it's great that "all multi-modal data" will be ingested in "realtime" into a "single event bus", the paper doesn't dive into the super granular technicals of how that data gets collected, streamed, checked for quality, or specifically governed."

Feedback: "Bias/Drift Fixes: The paper notes the importance of "bias and drift mitigation" for models. However, it doesn't lay out the specific technical methods or tools they'll use to actually tackle those issues."

Ziyad's Response:

Data Pipeline: Source -> Bus -> Trust -> Use

- **Source Capture:**
 - **Change-Data-Capture (CDC):** Debezium to Kafka
 - **Edge Sensors / CCTV:** RTSP -> GStreamer -> NATS JetStream
 - **Mobile & Web Events:** Snowplow -> Kafka
- **Ingestion Gate:**
 - **Kafka cluster** (with Rack & AZ aware partition placement)
 - **Topic Naming** = **<modality>.<domain>.<granularity>**
 - **Schema Registry (Apicurio)** -- Protobuf w/ backward-compat windows
- **Contract & Quality Firewall:**
 - **Git-backed great expectations checks** (null %, range, referential)
 - **ML Drift:** Kolmogorov - Smirnov on key features ($p < 0.01$ triggers quarantine)
 - **Quarantine Topics** auto-fork; Slack SRE alert
- **Governance & Lineage:**
 - **DataHub metadata to OpenLineage** (hooks emitted from Spark/Flink jobs)
 - **PII Classifier** (Presidio + custom regex) tags fields; row-level AES-GCM

- **Stream Processing:**
 - **Apache Flink (K8s session cluster)** performs multimodal joins feature enrichment (GeoIP, SKU risk score) and fan-out to “expert” topics for MoE core
- **Storage Tiers:**
 - **Hot(7 days):** Tier-1 Kafka + RocksDB state
 - **Warm (90 days):** Apache Iceberg on S3, Z-ORDER (modality, timestamp)
 - **Cold (1 yr):** Glacier Deep Archive, catalog in DataHub
- **Access & Privacy Gates:**
 - **Row-level filter in Flink SQL** (gdpr_delete = false)
 - **Tokenisation Service** (Format-Preserving AES) for MoE inference path
 - **Audit-Trail:** Immutable append-only LedgerDB

Drift Detection and Rectification Workflow Example

1. Ingestion monitors fire when, e.g., CCTV night-shift frames for one region fall below 2% of batch; the PSI alarm autotrigger a re-weigh & active-label job.
2. Fine-tuning loop picks up re-weighted shards; FairMix loss penalizes worst-slice AUC, and LoRA heads are updated within 5 min.
3. Router fairness penalty ensures expert cells aren’t only selected for “majority” geographies, avoiding latency bias.
4. Online drift (Deep-MMD p-value < 0.01) spins a shadow model; once canary shows TPR-gap <= 0.5 pp and P99 latency < baseline, it hot-swaps.
5. Audit pipeline writes model-card “Bias & Drift” with slice metrics; DataHub lineage links data versions, LoRA deltas, and gating-weights commit.

5. Knowledge-Graph (KG) Mechanics

Feedback: “Knowledge Graph Mechanics: The idea of an AI generating answers from a “knowledge graph” for explainability is cool. But we don’t hear much about the technical build, population, or query mechanisms of this graph itself.”

Ziyad’s Response:

Build/Logical Schema Property-Graph -- five top-level ontologies mirror the core MoE cells:

- Key vertex types: :Actor, :Asset, :Event, :Policy, :RiskSignal
- Every edge stores valid_from, valid_to, confidence, provenance_hash, enabling bitemporal queries and lineage

Population

- **Ingest:** Kafka to Flink enables X GB/s streaming ingest from POS, CCTV, audit logs, etc.
- **Pre-processing:** HuggingFace NER+RE + CLIP enables extracting entities / relations from text & vision
- **Canonicalise:** MinHash + UF Merge-Find enables deduplication across sources
- **Linkage:** Probabilistic Soft Logic matches to existing vertices; emit sameAs edges with score
- **Enrich:** GraphSAGE Embeddings give 256-dim vector per node; stored in Faiss for ANN joins
- **Write:** JanusGraph batch-mutator handles 50k edges/s per shard, idempotent upserts

Query

- **User / Upstream Service**
 - natural-language question
- **Query Rewriter (T5-XXL)**
 - predicts KG-aware SPARQL / Gremlin template
 - injects domain + modality constraints from MoE router
- **KG Query Planner**
 - cost-based plan: (RedisGraph -> JanusGraph -> Faiss fallback)
- **Execution Engine**
 - gathers sub-graph, introspects edge confidences

- **Evidence Assembler**
 - ranks paths with GNN scorer + path-length penalty
 - selects ≤ 3 "explanation slices"
- **Answer Generator (LoRA head)**
 - formats natural-language answer
 - embeds citation hooks pointing to slice IDs
- **Response (and clickable graph visualization)**
 - Single-page react app with graph visualizer + websockets

6. Compute & Distributed Training Footprint

Feedback: "Broader Compute Infrastructure: It mentions "serverless architecture" for agentic workflows, which is a good start. But for the whole massive "Walmart AI" system, it doesn't really touch on the broader computational infrastructure needed, like specific compute resources or distributed training frameworks."

Ziyad's Response:

Walmart AI Fabric Capacity Planning

- **Core Training:** 3 x 640-GPU rings (NVIDIA H100 80 GB w/ NVSwitch), 2 x 512-TPU v5e pods with 400 Gb InfiniBand NDR + NVLink4 interconnects
- **Fine-Tuning:** 6 x 128-GPU DGX-GH200 partitions (Grace Hoppers) with 200 Gb RoCEv2 interconnects
- **Realtime Serving:** 1200 x H100 SXM GPUs (8-GPU DGX-H100 nodes) across 4 AZs with 100 Gb EFA
- **Edge Storage:** 2-GPU L4 / H100 PCIe per SuperMicro node in 450 stores with 25 Gb Ethernet

Distributed Training Stack

1. **Framework** -> PyTorch 2.3 + TorchInductor
2. **Scaffolding** -> DeepSpeed 0.14 + Megatron-LM adapters for 3D parallelism (PP=6, TP=8, DP=16)
3. **Optimizer Offload** -> FSDP-ZeRO-3 with NVMe offload (8 x 7 TB PCIe 4.0 per node)
4. **KV-Cache & Router** -> FlashAttention-2 + tensor parallel rotary cache sharding
5. **Experiment Control** -> Weights & Biases + MLflow lineage -> DataHub metadata
6. **Checkpoint Format** -> Safetensors + EcPh Zoo (60 GB/s parallel HDFS restore)
7. **Elastic Fault-Tolerance** -> TorchElastic over Slurm restart-groups; 8 node failure budget before job abort